

Behind the Streams: Twitch's Technical Design

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Introduction

Twitch is a leading live-streaming platform that uses sophisticated server systems and machine-learning algorithms to deliver seamless, real-time content.

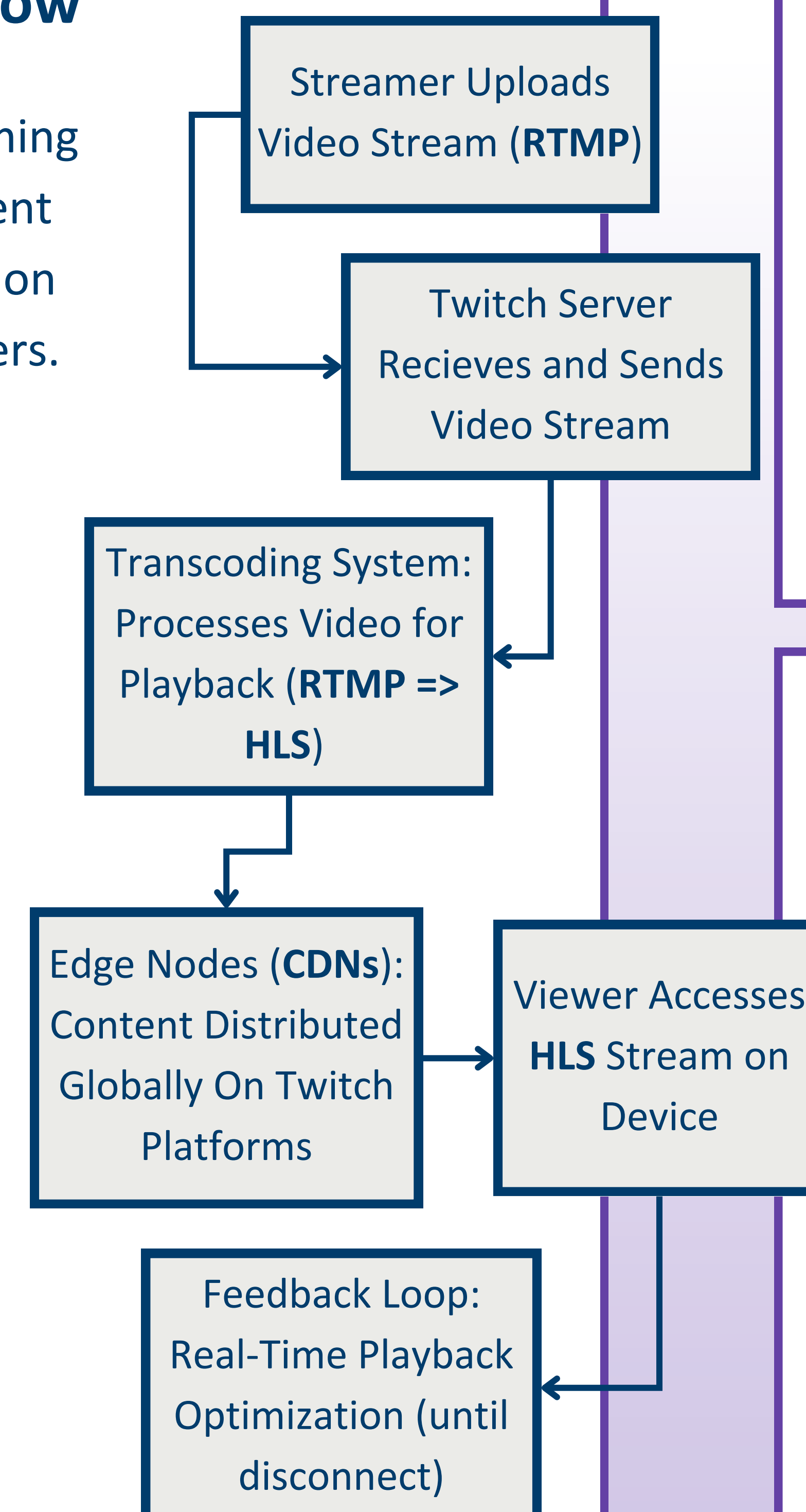
These mechanisms enable Twitch to:

- Provide personalized watching recommendations.
- Minimize latency for real-time interactions.
- Support millions of users globally as content creators and watchers.

Server Flow

Delivering seamless live-streaming content depends on an efficient system to bridge the connection between streamers and viewers.

- **Real-Time Messaging Protocol (RTMP):** A protocol delivering raw video and audio data to a destination
- **HTTP Live Streaming (HLS):** Format which breaks video data into smaller segments for adaptability
- **Content Delivery Networks (CDNs):** Global network of servers designed to replicate and deliver content efficiently
- **Playback:** The process where viewers receive and watch content



Recommendation Systems: Tailoring Content to Viewers

Inputs:

- Interests Survey
- Watch History
- User Engagements
 - Interactions on channels and categories
 - Likes, Dislikes, Time Spent
- User Preferences
 - Language, Region

Processing Algorithm:

Machine Learning Model + Suggestion Strategies

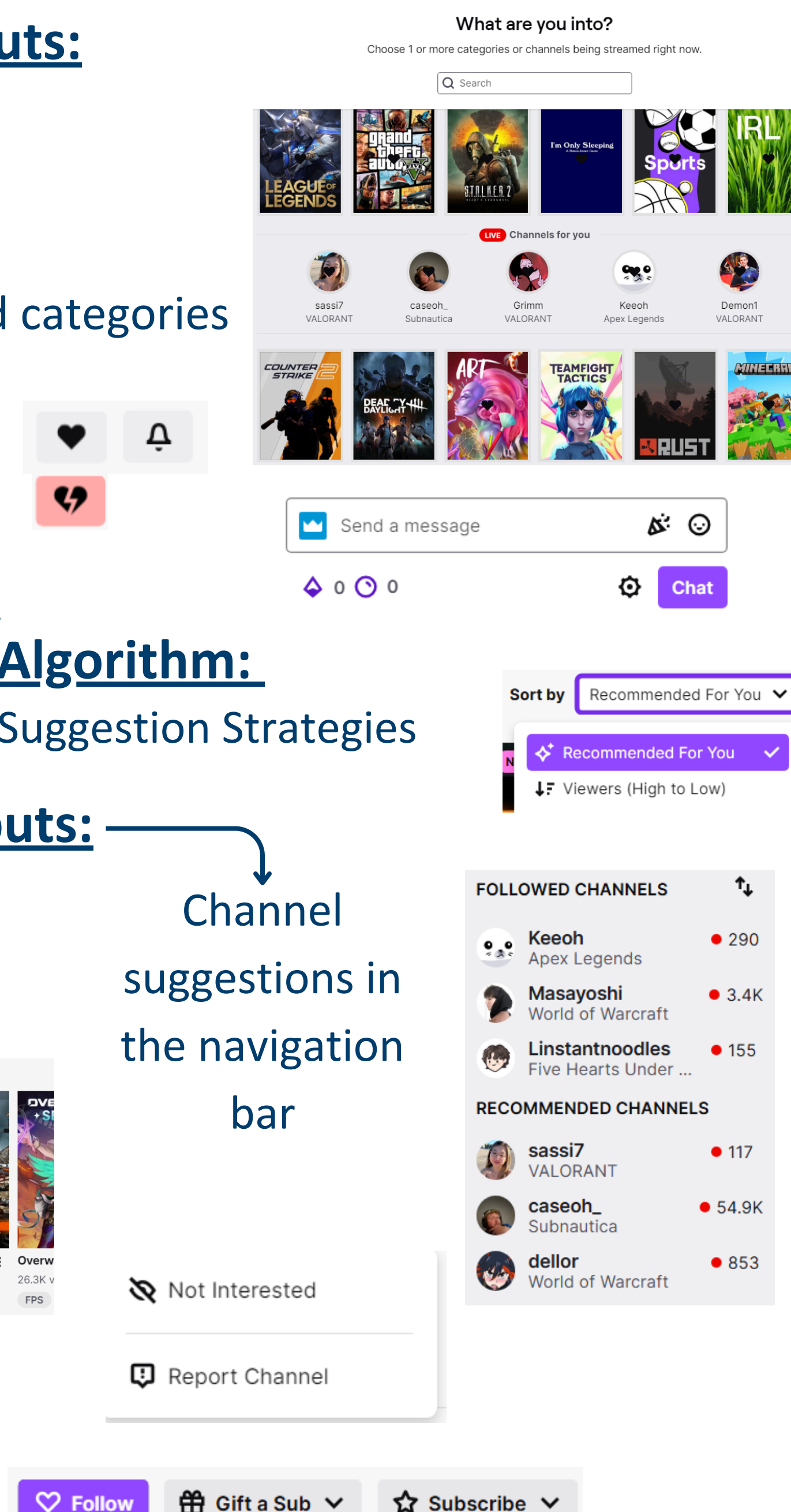
Outputs:

Curated section of suggested on-going streams and categories

Channel suggestions in the navigation bar

User Fine-Tuning:

- Continued engagements
- “Not Interested” button feature



Latency Management: Real-Time Interaction Maintenance

Latency: Time Delays between an action occurring and its corresponding response

- Where it occurs:
 - between “Streamer Uploads Video Stream” and “Twitch Server Receives and Sends Video Stream”
 - when “Viewer Accesses HLS Stream on Device”
- Goal: Low Latency = minimized delays

Mechanisms Involved:

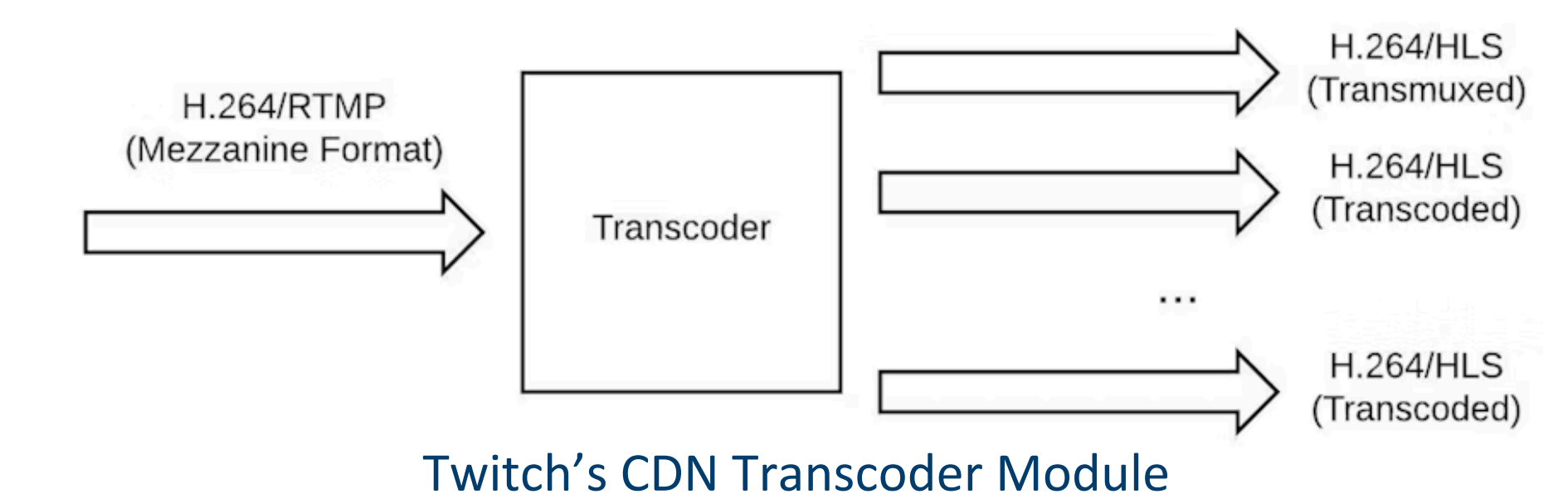
- **Bitrate Optimization:** Video quality stabilization
- **Adaptive Bitrate Streaming:** Dynamic quality adjustment to internet speed
- Content Delivery Networks (CDNs): Global distribution
- **Efficient Encoding/Decoding:** Fast video compression (NVIDIA NVENC, x264)
- **Edge Nodes:** Handles local traffic for faster data traveling

Compression Techniques

Compression: making video streams adaptive across all networks

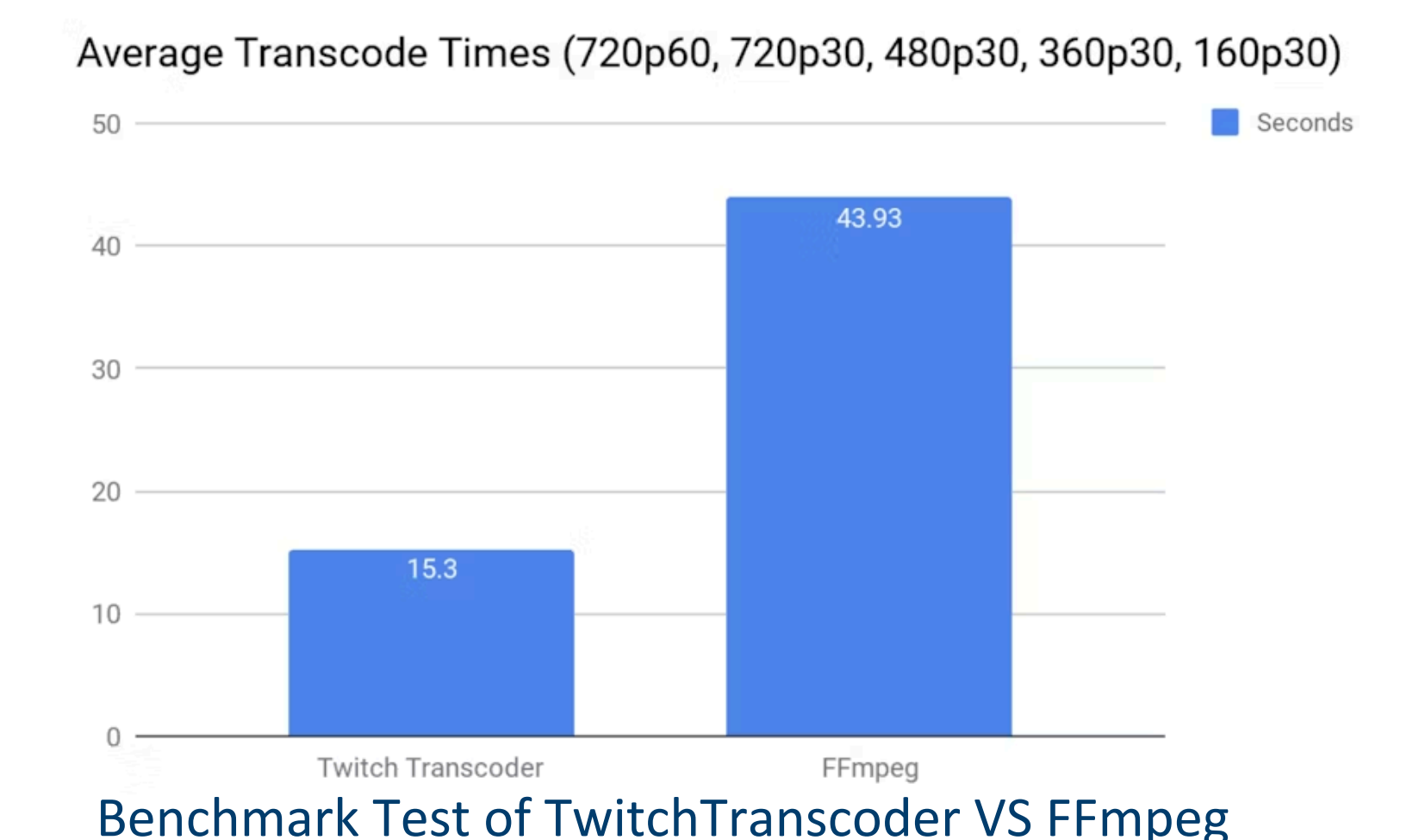
Key Technologies involved: [6,7]

- **TwitchTranscoder:** multithreaded optimization
 - custom-built software to process thousands of simultaneous streams
- **Adaptive BitRate Ladder:** (RTMP => HLS) creating multiple resolution options
- **H.264 Codec:** Balances file size with quality retention



Efficiency: [7]

- Optimized resource is **65%** faster in execution compared to FFmpeg, popular software



Citations

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- [4] “Twitch Help Portal,” help.twitch.tv. https://help.twitch.tv/s/article/broadcasting-guidelines?language=en_US
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- [6] J. Gong, S. Dhanju, C.-C. Lu, and Y. Shen, “Live Video Transmuxing/Transcoding: FFmpeg vs TwitchTranscoder, Part I,” Twitch.tv, Oct. 10, 2017. <https://blog.twitch.tv/en/2017/10/10/live-video-transmuxing-transcoding-f-ffmpeg-vs-twitch-transcoder-part-i-489c1c125f28/> (accessed Nov. 17, 2024).
- [7] J. Gong, S. Dhanju, C.-C. Lu, and Y. Shen, “Live Video Transmuxing/Transcoding: FFmpeg vs TwitchTranscoder, Part II,” Twitch.tv, Oct. 23, 2017. <https://blog.twitch.tv/en/2017/10/23/live-video-transmuxing-transcoding-f-ffmpeg-vs-twitch-transcoder-part-ii-4973f475f8a3/>